

Generative Modelling with GANs: From Synthetic Distributions to Real-World and Creative Applications

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GitHub: <https://github.com/giridharreddygoddilla/Generative-Modelling-Case-Study>

1. Introduction

Generative modelling has become an important area in modern data science, particularly for synthetic data generation, data augmentation, and privacy-preserving analytics. Among generative approaches, Generative Adversarial Networks (GANs) have shown strong ability to learn complex, high-dimensional data distributions without explicit likelihood estimation (Goodfellow et al., 2014). Unlike traditional probabilistic models, GANs learn distributions implicitly through adversarial training, which makes them flexible but also difficult to optimise.

This project investigates GAN performance across three progressively complex domains:

- synthetic low-dimensional data (sine wave and spiral),
- structured real-world tabular data (CICIDS cybersecurity dataset),
- real-world image datasets (BloodMNIST and QuickDraw sketches).

The central objective is to evaluate how effectively GANs learn and reproduce data distributions of increasing complexity, and to identify the limitations that arise across different data modalities.

By comparing performance across these settings, the study highlights both the strengths and the practical limitations of GAN-based generative modelling.

2. Methodology

2.1 Experimental Design

The study is organised into four stages:

Part	Dataset Type	Objective
Part 1	Toy data	Understand GAN behaviour
Part 2.1	Medical images	Evaluate realism
Part 2.2	Tabular data	Test structured data modelling
Part 2.3	Sketch images	Analyse simpler visual domain

This staged design creates a controlled increase in data complexity, allowing systematic comparison across dimensionality, structure, and modality.

2.2 GAN Framework and Training

A standard GAN framework was used:

- **Generator:** maps latent vector $z \sim \mathcal{N}(0,1)$ to data space
- **Discriminator:** binary classifier distinguishing real and generated samples

Training follows a minimax objective in which the generator attempts to produce samples indistinguishable from real data, while the discriminator improves its classification accuracy.

GAN training is inherently unstable because of:

- non-convex optimisation,

- lack of guaranteed equilibrium,
- mode collapse, where the generator produces limited diversity.

These issues become more severe for high-dimensional or complex datasets, as shown in the later experiments.

2.3 Architecture Design Choices

Data Type	Architecture	Justification
Toy Data	MLP	Low-dimensional
BloodMNIST	DCGAN	Spatial structure
CICIDS	MLP GAN	Tabular features
QuickDraw	DCGAN	Image-Like-Sketches

DCGAN (Radford et al., 2016) was used for image-based tasks because convolutional layers:

- preserve spatial locality,
- capture hierarchical features such as edges and textures,
- improve gradient flow and training stability.

In contrast, CICIDS is tabular and has no spatial structure, so a fully connected GAN was more appropriate.

2.4 Evaluation Strategy

Data Type	Evaluation
Toy	Visual + statistics
Images	FID
Tabular	PCA + distribution comparison

The Fréchet Inception Distance (FID) measures similarity between real and generated feature distributions in deep feature space, where lower values indicate closer alignment (Heusel et al., 2017).

Evaluation methods were chosen according to data modality to avoid drawing conclusions from unsuitable metrics.

3. Part 1: Toy Data Experiments

3.1 Sine Wave

Sine-wave summary:

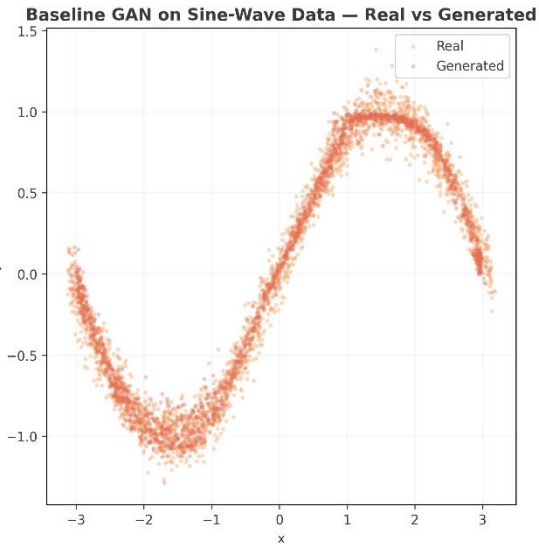
	experiment	metric	real	generated
0	Baseline GAN on Sine-Wave Data	mean_x	-0.008564	-0.074429
1	Baseline GAN on Sine-Wave Data	mean_y	0.010462	0.004920
2	Baseline GAN on Sine-Wave Data	std_x	1.835556	1.834331
3	Baseline GAN on Sine-Wave Data	std_y	0.712926	0.713843

Analysis

The GAN successfully learns the sine-wave manifold. The generated samples follow the continuous function closely and align strongly with the real data in both shape and spread.

Key observations include:

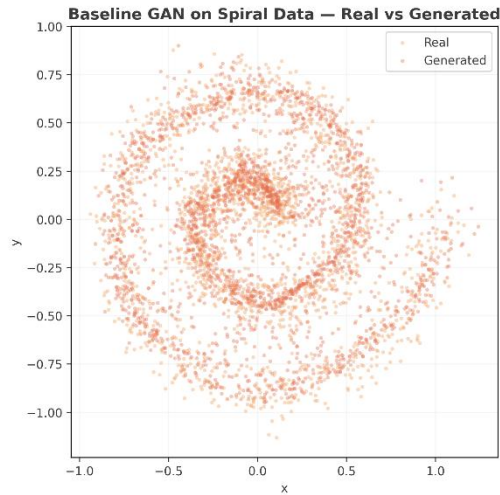
- near-perfect variance alignment,
- minimal deviation around peaks and troughs,
- a slight horizontal shift indicating a small phase offset.



	absolute_difference
0	0.065865
1	0.005542
2	0.001225
3	0.000917

This shows that the model captures both the underlying function and the noise around it.

3.2 Spiral - Baseline GAN



Spiral baseline summary:					absolute_difference
	experiment	metric	real	generated	
0	Baseline GAN on Spiral Data	mean_x	-0.002738	0.006012	0.008749
1	Baseline GAN on Spiral Data	mean_y	-0.073826	-0.057837	0.015989
2	Baseline GAN on Spiral Data	std_x	0.437880	0.440161	0.002281
3	Baseline GAN on Spiral Data	std_y	0.428805	0.414769	0.014036

Analysis

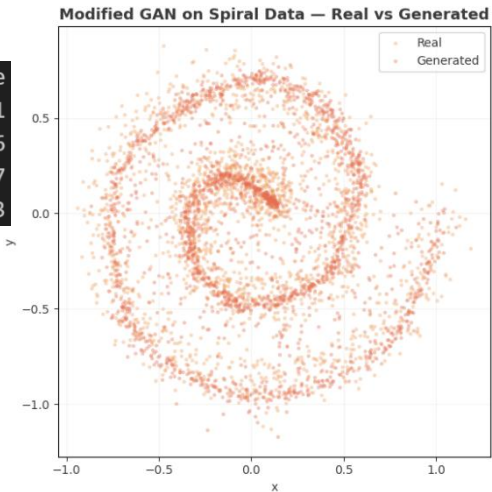
Although the summary statistics suggest close alignment, the generated samples fail to reproduce the spiral structure convincingly:

- points are scattered across the space,
- spiral arms are weak,
- geometric coherence is limited.

This shows an important limitation: GANs can match global statistics without learning the actual manifold structure.

3.3 Spiral — Modified GAN

Spiral modified summary:					absolute_difference
	experiment	metric	real	generated	
0	Modified GAN on Spiral Data	mean_x	-0.002738	0.024543	0.027281
1	Modified GAN on Spiral Data	mean_y	-0.073826	-0.095062	0.021236
2	Modified GAN on Spiral Data	std_x	0.437880	0.428272	0.009607
3	Modified GAN on Spiral Data	std_y	0.428805	0.458638	0.029833



Analysis

The modified architecture improves the spiral outputs through:

- clearer spiral arms,
- better spatial organisation,
- less diffuse noise.

However, the simple statistical summaries become slightly worse than the baseline. This indicates that structural realism is not always captured by basic statistical measures.

Conclusion (Part 1)

The baseline spiral GAN provides a better statistical match, while the modified GAN produces a stronger structural representation. Overall, Part 1 shows that visual and quantitative evaluation must be combined.

4. Part 2.1 - BloodMNIST

4.1 Blood dataset overview

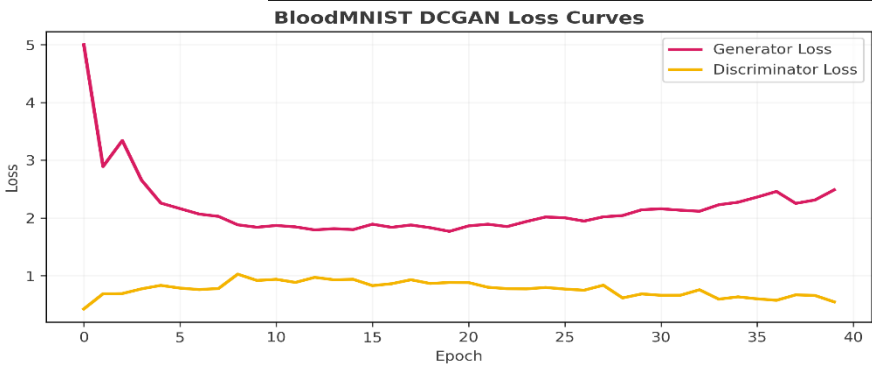
The dataset contains **11,959 training images** across 8 classes, with noticeable class imbalance.

	split	num_images	image_height	image_width	channels
0	train	11959	28	28	3
1	validation	1712	28	28	3
2	test	3421	28	28	3

4.2 Training Behaviour

Final losses:

- Generator: **2.49**
- Discriminator: **0.55**



Analysis

Training follows typical GAN behaviour:

- early improvement,
- generator loss increasing after about 20 epochs,
- discriminator gradually becoming stronger.

This suggests that the discriminator learns faster and eventually reduces the quality of the learning signal available to the generator.

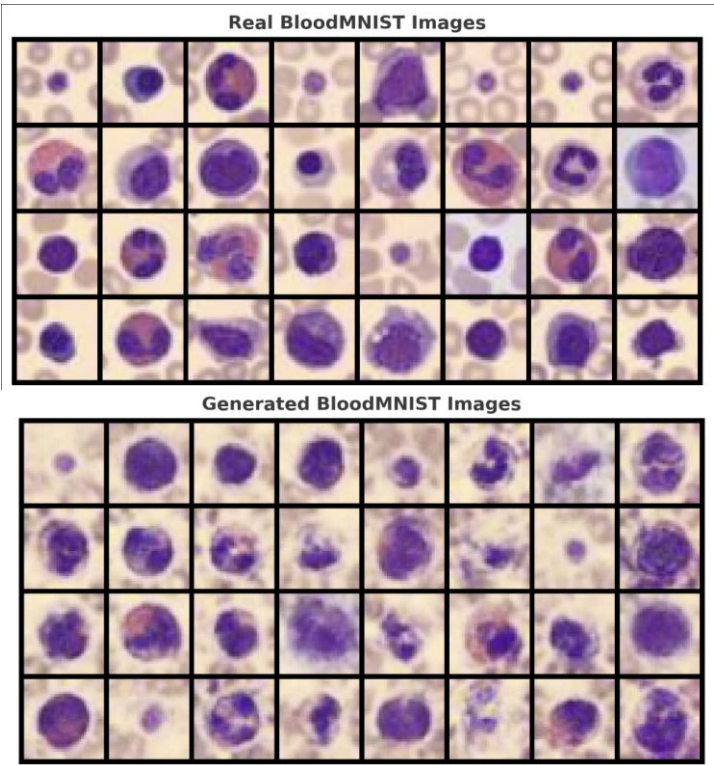
4.3 Generated Image Quality

Generated samples show:

- broadly correct colour distributions,
- approximate cell-like shapes,
- nucleus-like purple regions.

However:

- fine texture is missing,
- edges are blurred,
- several outputs remain biologically unconvincing.



4.4 Quantitative Evaluation

- **FID = 210.65**

Interpretation

BloodMNIST FID: 210.6532					
	metric	value	num_real_images	num_fake_images	status
0	FID	210.653183	1000	1000	computed_successfully

This high FID value indicates a large gap between real and generated image distributions. In practice, this means the model captures coarse visual appearance but not high-level feature realism.

Conclusion

The BloodMNIST GAN learns basic visual structure but fails to generate high-fidelity medical images. This is consistent with previous work showing that biomedical texture is difficult to synthesise accurately (Frid-Adar et al., 2018).

5. Part 2.2 - CICIDS (Tabular Data)

5.1 Dataset Overview

The filtered dataset contains approximately **691,395 samples** and **68 numerical features**, representing BENIGN and DoS traffic flows. These features include packet counts, flow durations, and byte-level statistics, many of which are strongly skewed and interdependent.

Unlike images, these features do not contain spatial relationships, which makes them harder to model using standard GANs.

	item	value
0	rows_used_for_training	691395
1	num_features_used	68
2	benign_rows_used	439683
3	dos_rows_used	251712
4	epochs	30
5	batch_size	256
6	latent_dim	64
7	final_generator_loss	2.261579
8	final_discriminator_loss	0.548451
9	pca_sample_size	5000
10	device_used	cpu

5.2 Distribution Characteristics

The real CICIDS data shows several difficult properties:

- heavy-tailed distributions such as Flow Duration,
- extreme outliers spanning multiple orders of magnitude,
- multi-modal patterns across benign and attack traffic,
- strong feature correlations, for example between packet counts and packet lengths.

These characteristics violate the smooth, relatively well-behaved distributions that standard GANs tend to learn more easily.

5.3 Generated vs Real Feature Comparison

The generated data differs substantially from the real distribution.

For example:

- **Flow Duration**
 - Real mean: 2.8×10^7
 - Generated mean: 7.1×10^6

This represents approximately a **75% underestimation**, indicating that the GAN fails to reproduce the long-tail behaviour of the distribution.

Other features show a similar pattern, where packet-based variables exhibit significantly lower variance, generated values are overly concentrated around central regions, and rare extreme values are largely absent.

Interpretation

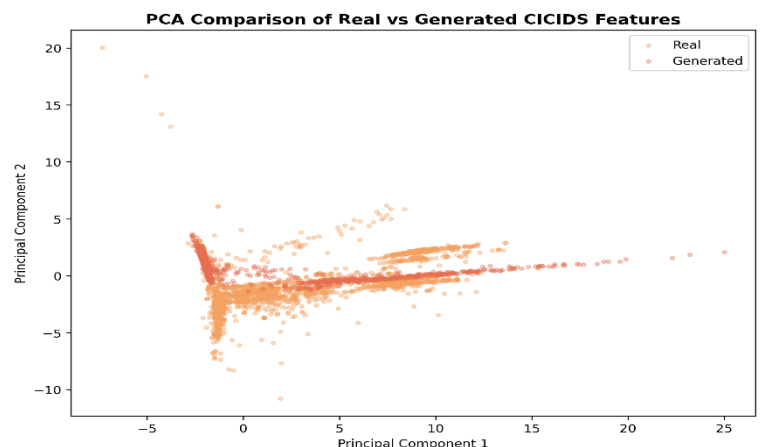
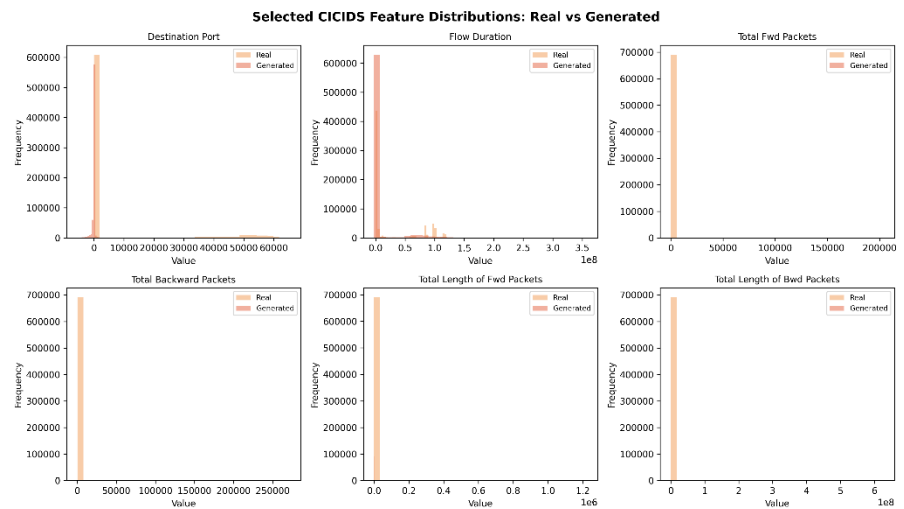
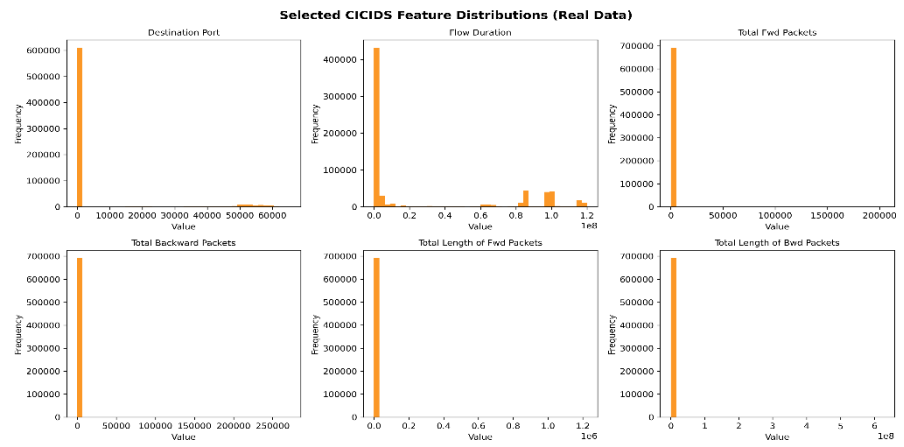
This behaviour is consistent with *mode averaging*, where the generator learns dominant high-density regions while ignoring rarer parts of the distribution. As a result, the synthetic traffic appears smoother and less diverse than the real data. This limitation is particularly critical in cybersecurity contexts, where rare or extreme patterns often correspond to attack behaviour, reducing the practical usefulness of the generated data.

5.4 PCA-Based Distribution Analysis

The PCA projection gives a low-dimensional view of the learned feature space.

The main observations are:

- real data forms wide, dispersed clusters,
- generated data is more compressed,
- some overlap exists, but large parts of the real space remain uncovered.



Interpretation

The generator captures the central mass of the distribution but fails to reproduce:

- sparse regions,
- boundary behaviour,
- high-variance directions.

This suggests incomplete learning of the real traffic manifold.

5.5 Deep Technical Explanation

The weak CICIDS performance can be explained by several technical limitations.

1. High Dimensionality

With **68 features**, the generator must learn a complex joint distribution:

$$P(x_1, x_2, \dots, x_{68})$$

This makes accurate distribution learning substantially harder than in low-dimensional toy settings.

2. Feature Dependency Structure

Unlike images, where neighbouring pixels have strong local relationships, tabular data contains:

- non-linear and non-local dependencies,
- heterogeneous feature scales.

Standard GANs do not explicitly model these relationships well.

3. Heavy-Tailed Distributions

Many CICIDS features are long-tailed. GAN training tends to prioritise dense regions of the data space, so rare extreme values contribute very little to optimisation.

4. Mode Collapse

The generator may collapse to a limited subset of modes, producing:

- repetitive samples,
- reduced diversity,
- variance compression.

This is supported by both the histogram comparisons and the PCA plot.

5.6 Conclusion (Part 2.2)

The results show that standard GANs are poorly suited to tabular data generation when:

- feature distributions are highly skewed,
- dependencies are complex,
- rare events matter.

This is consistent with prior work suggesting that specialised models such as CTGAN (Xu et al., 2019) are more appropriate for realistic tabular synthesis.

6. Part 2.3 - QuickDraw Pizza (Creative AI)

6.1 Dataset Characteristics

	dataset	num_images	image_height	image_width	channels
0	QuickDraw Pizza	130371	28	28	1

The dataset contains **130371 grayscale pizza sketches**, each represented as a **28 × 28** image. Compared with BloodMNIST, these sketches are:

- highly sparse,
- structurally simple,
- visually consistent.

6.2 Training Dynamics

Final losses:

- Generator: **3.40**
- Discriminator: **0.46**

Analysis

The training pattern follows a common GAN dynamic:

- the discriminator becomes highly confident,
- generator loss increases as it tries to fool a stronger discriminator.

However, unlike BloodMNIST, the visual quality continues to improve, showing that generator loss alone is not a reliable measure of perceptual quality.

6.3 Visual Progression of Generated Samples

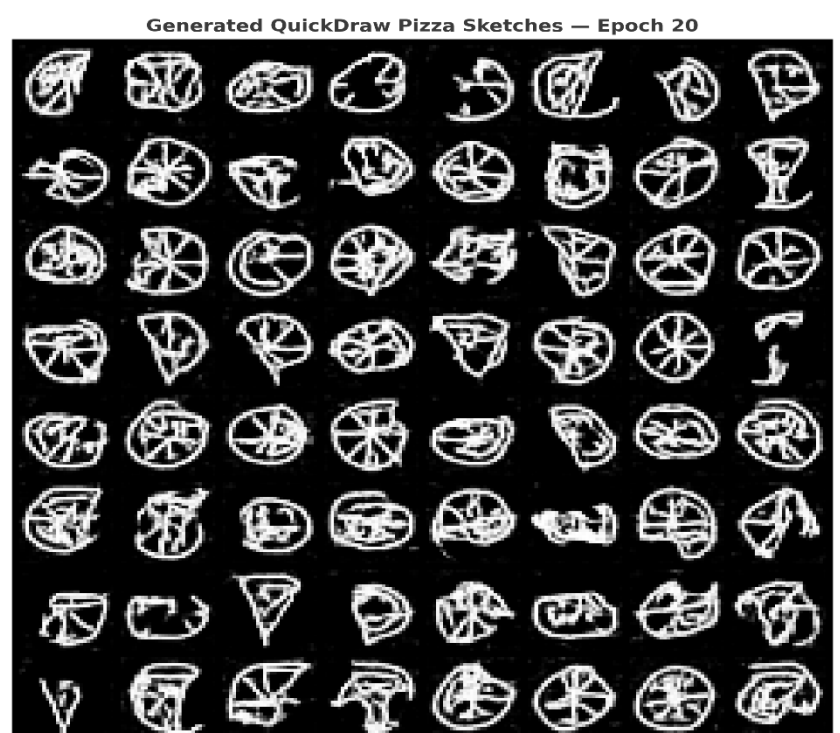
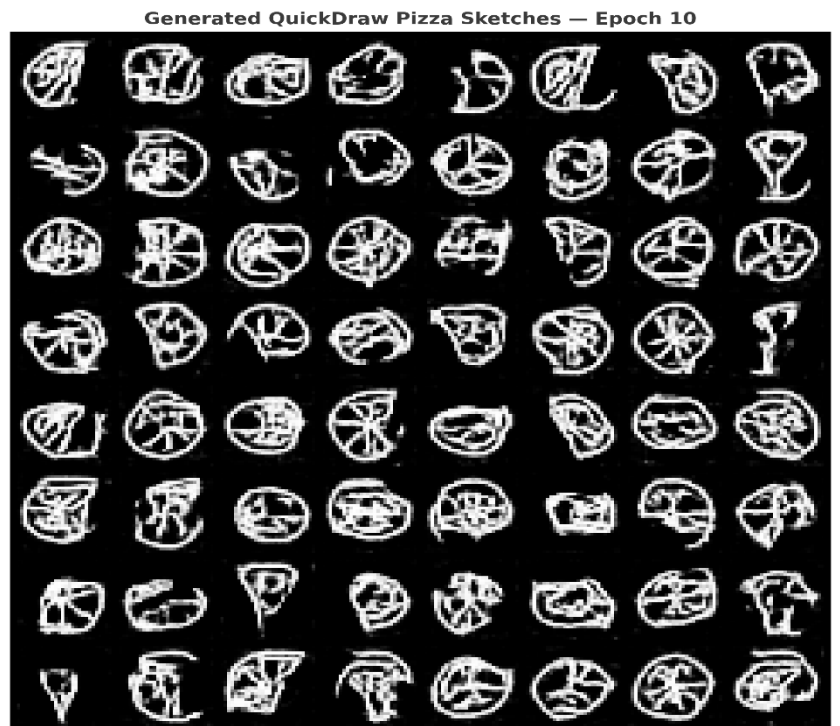
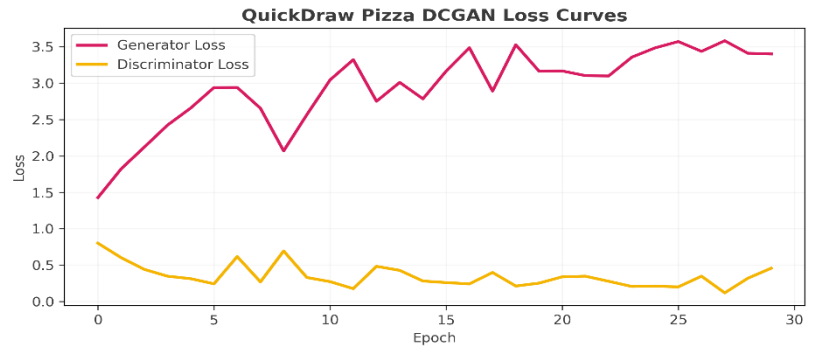
The generated samples improve clearly over training:

- early epochs produce noise-like images,
- mid training produces circular outlines,
- late training produces recognisable pizza shapes with slice-like segmentation.

Interpretation

This progression suggests that the generator learns:

- global structure first,
- then more detailed internal patterns.



That hierarchical pattern is consistent with convolutional feature learning. **6.4 Final Output Quality**

The final generated sketches capture:

- circular pizza geometry,
- radial segmentation,
- overall sketch consistency.

The remaining issues are:

- broken or incomplete lines,
- inconsistent slice definition,
- occasional distortions.

6.5 Quantitative Evaluation

- **FID = 41.75**

QuickDraw Pizza FID: 41.7457				
metric	value	num_real_images	num_fake_images	status
0 FID	41.745682	1000	1000	computed_successfully

Interpretation

Compared with BloodMNIST (FID $\approx$ 210.55), this represents a substantial improvement of over 80%, indicating significantly better alignment with the real data distribution.

6.6 Why QuickDraw Works Better

The stronger performance can be explained by three factors.

- **Lower Complexity:** The shapes are simple and follow consistent visual rules.
- **Reduced Variability:** There is less intra-class diversity than in medical images.
- **Sparse Representation:** The sketches contain fewer active pixels, which reduces the number of dependencies the model must learn.

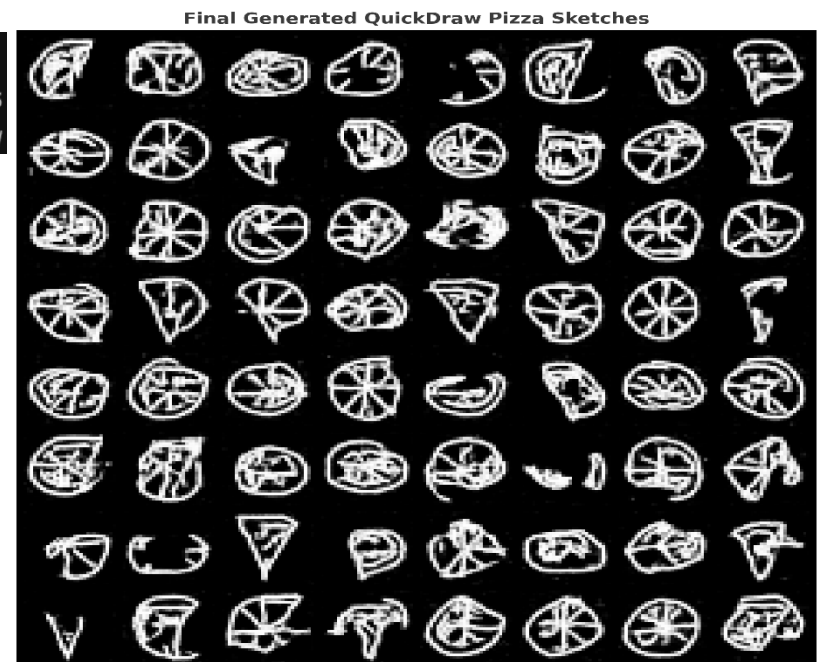
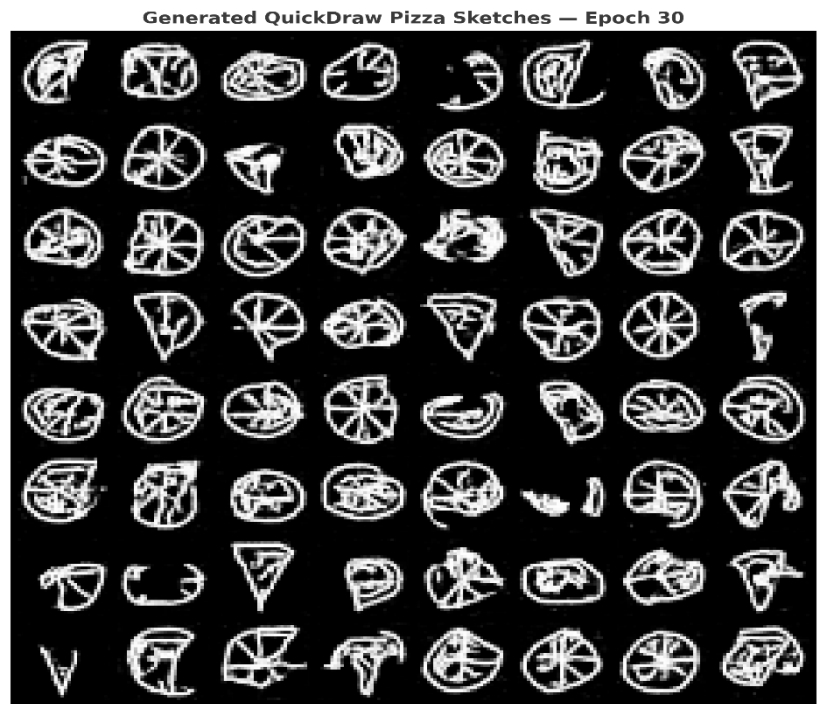
6.7 Conclusion (Part 2.3)

The QuickDraw experiment shows that GANs perform best when: **Structure is simple, intra-class variability is limited, the visual domain is consistent.**

7. Discussion

7.1 Impact of Data Modality

The experiments show that GAN performance depends strongly on data modality, with highly effective learning on continuous low-dimensional data, moderate success on image data, and significantly weaker performance on tabular data due to its structural complexity. These findings are consistent with existing research, which shows that GANs perform best when the underlying data has strong spatial structure and lower intrinsic dimensionality, while performance degrades in high-dimensional tabular settings.



7.2 Key Limitations of GANs

Mode collapse is particularly evident in the CICIDS results, where the generated data lacks the **diversity and variance** observed in real samples. **Training instability** is also observed in image-based experiments such as BloodMNIST and QuickDraw, where **discriminator dominance** leads to increasing generator loss. More broadly, GANs struggle to model distributions that are **heavy-tailed, multi-modal, and high-dimensional**, limiting their ability to fully represent real-world data.

7.3 Evaluation Considerations

A major finding of this study is that no single evaluation metric is sufficient. Statistical summaries can be misleading, visual inspection remains essential for assessing realism, and while FID provides a useful quantitative measure for image data, it does not fully capture all aspects of generative quality. This highlights the need for a multi-faceted evaluation approach, where quantitative metrics are complemented by qualitative analysis to provide a more reliable assessment of model performance.

8. Conclusion

GAN performance is highly sensitive to data structure, feature complexity, and architecture choice.

The results show strong performance on simple synthetic distributions, moderate success on structured image data, and clear limitations on tabular datasets.

Overall, this highlights the importance of matching architectures to the data type, combining qualitative and quantitative evaluation, and recognising the inherent limits of adversarial training.

These findings also emphasise that model performance should be interpreted alongside data characteristics, rather than relying solely on numerical metrics.

In practice, selecting the right modelling approach is as important as the model itself for achieving reliable generative performance.

Furthermore, the study demonstrates that improvements in model design alone are insufficient without considering the underlying data distribution and its complexity.

This reinforces the need for a data-centric approach to generative modelling, where both model selection and data characteristics are jointly optimised.

9. Future Work

Several extensions could further improve performance and robustness.

Conditional GANs could enable class-controlled generation and improve output diversity.

WGAN-GP could enhance training stability by enforcing Lipschitz continuity (Gulrajani et al., 2017).

Tabular-specific models such as CTGAN are better suited for capturing feature dependencies, multi-modality, and skewed distributions.

Higher-resolution architectures such as StyleGAN could improve texture realism and fine-grained visual detail.

Future work could also explore cross-domain generalisation by training across multiple datasets and evaluating the transferability of learned representations.

Additionally, incorporating more robust evaluation techniques could provide a deeper understanding of generative quality across different data modalities.

Exploring hybrid approaches that combine GANs with probabilistic or diffusion-based models may further improve stability and realism.

A more rigorous evaluation framework, combining multiple metrics with human-level assessment, would also strengthen the reliability of future generative models.

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